a)

```
mysql> select * from employees order by last_name asc limit 10;
+--------+------------+------------+-----------+--------+------------+
| emp_no | birth_date | first_name | last_name | gender | hire_date  |
+--------+------------+------------+-----------+--------+------------+
|  11761 | 1964-07-17 | Bartek     | Aamodt    | M      | 1991-06-12 |
|  15427 | 1959-03-06 | Aluzio     | Aamodt    | M      | 1985-03-03 |
|  18182 | 1963-02-23 | Dekang     | Aamodt    | F      | 1988-05-25 |
|  16572 | 1956-11-26 | Matt       | Aamodt    | M      | 1987-06-16 |
|  12791 | 1960-06-16 | Mokhtar    | Aamodt    | M      | 1994-08-14 |
|  12516 | 1958-06-25 | Sreenivas  | Aamodt    | F      | 1990-03-06 |
|  12982 | 1952-12-08 | Sachem     | Aamodt    | F      | 1992-01-11 |
|  17400 | 1962-03-22 | Basim      | Aamodt    | F      | 1991-09-15 |
|  19898 | 1957-03-09 | Vidar      | Aamodt    | M      | 1988-08-06 |
|  17885 | 1954-02-01 | Takanari   | Aamodt    | M      | 1996-08-19 |
+--------+------------+------------+-----------+--------+------------+
10 rows in set (0.18 sec)

mysql>
```

b)

```
mysql> select * from employees order by last_name asc, first_name asc limit 10;
+--------+------------+------------+-----------+--------+------------+
| emp_no | birth_date | first_name | last_name | gender | hire_date  |
+--------+------------+------------+-----------+--------+------------+
| 258641 | 1961-05-23 | Abdelkader | Aamodt    | M      | 1994-12-02 |
| 258005 | 1953-02-17 | Adhemar    | Aamodt    | F      | 1991-01-21 |
| 455773 | 1960-05-04 | Aemilian   | Aamodt    | M      | 1988-04-21 |
| 436560 | 1959-03-16 | Alagu      | Aamodt    | F      | 1991-10-17 |
| 266651 | 1959-05-28 | Aleksander | Aamodt    | F      | 1989-03-29 |
| 487598 | 1962-03-03 | Alexius    | Aamodt    | M      | 1994-12-30 |
| 216963 | 1960-07-16 | Alois      | Aamodt    | M      | 1995-08-24 |
|  15427 | 1959-03-06 | Aluzio     | Aamodt    | M      | 1985-03-03 |
| 100860 | 1964-06-20 | Amabile    | Aamodt    | F      | 1993-02-06 |
| 107070 | 1954-04-24 | Anestis    | Aamodt    | M      | 1990-10-30 |
+--------+------------+------------+-----------+--------+------------+
10 rows in set (0.29 sec)

mysql>
```

c)

```
mysql> select * from employees order by hire_date desc limit 5;
+--------+------------+------------+-----------+--------+------------+
| emp_no | birth_date | first_name | last_name | gender | hire_date  |
+--------+------------+------------+-----------+--------+------------+
| 463807 | 1964-06-12 | Bikash     | Covnot    | M      | 2000-01-28 |
| 428377 | 1957-05-09 | Yucai      | Gerlach   | M      | 2000-01-23 |
| 499553 | 1954-05-06 | Hideyuki   | Delgrande | F      | 2000-01-22 |
| 222965 | 1959-08-07 | Volkmar    | Perko     | F      | 2000-01-13 |
|  47291 | 1960-09-09 | Ulf        | Flexer    | M      | 2000-01-12 |
+--------+------------+------------+-----------+--------+------------+
5 rows in set (0.15 sec)

mysql>
```

d)

```
mysql> select e.first_name, e.last_name, s.salary from employees e join salaries s on e.emp_no = s.emp_no order by s.sal
ary desc limit 1;
+------------+-----------+--------+
| first_name | last_name | salary |
+------------+-----------+--------+
| Tokuyasu   | Pesch     | 158220 |
+------------+-----------+--------+
1 row in set (2.20 sec)

mysql>
```

e)

```
mysql> select e.first_name, e.last_name, s.salary from employees e join salaries s on e.emp_no = s.emp_no order by s.sal
ary asc limit 1;
+------------+-----------+--------+
| first_name | last_name | salary |
+------------+-----------+--------+
| Olivera    | Baek      |  38623 |
+------------+-----------+--------+
1 row in set (2.59 sec)

mysql>
```

f)

```
mysql> select e.first_name, e.last_name, s.salary from employees e join salaries s on e.emp_no = s.emp_no where s.salary
 > 150000;
+------------+-----------+--------+
| first_name | last_name | salary |
+------------+-----------+--------+
| Tokuyasu   | Pesch     | 151115 |
| Tokuyasu   | Pesch     | 153166 |
| Tokuyasu   | Pesch     | 153458 |
| Tokuyasu   | Pesch     | 157821 |
| Tokuyasu   | Pesch     | 158220 |
| Ibibia     | Junet     | 150345 |
| Xiahua     | Whitcomb  | 151929 |
```

g)

```
mysql> select d.dept_name, count(de.emp_no) as tyontekijamaara from departments d join dept_emp de on d.dept_no = de.dep
t_no where d.dept_name in ('sales', 'marketing') group by d.dept_name;
+-----------+-----------------+
| dept_name | tyontekijamaara |
+-----------+-----------------+
| Marketing |           20211 |
| Sales     |           52245 |
+-----------+-----------------+
2 rows in set (0.07 sec)

mysql>
```

h)

```
mysql> select e.first_name, e.last_name, d.dept_name from employees e join dept_manager dm on e.emp_no = dm.emp_no join
departments d on dm.dept_no = d.dept_no;
+-------------+--------------+--------------------+
| first_name  | last_name    | dept_name          |
+-------------+--------------+--------------------+
| Tonny       | Butterworth  | Customer Service   |
| Marjo       | Giarratana   | Customer Service   |
| Xiaobin     | Spinelli     | Customer Service   |
| Yuchang     | Weedman      | Customer Service   |
| DeForest    | Hagimont     | Development        |
| Leon        | DasSarma     | Development        |
| Ebru        | Alpin        | Finance            |
| Isamu       | Legleitner   | Finance            |
| Shirish     | Ossenbruggen | Human Resources    |
| Karsten     | Sigstam      | Human Resources    |
| Margareta   | Markovitch   | Marketing          |
```

i)

```
mysql> select d.dept_name, avg(s.salary) as keskipalkka from departments d join dept_emp de on d.dept_no = de.dept_no j
oin salaries s on de.emp_no = s.emp_no where d.dept_name in ('sales', 'marketing') group by d.dept_name;
+-----------+-------------+
| dept_name | keskipalkka |
+-----------+-------------+
| Marketing |  71913.2000 |
| Sales     |  80667.6058 |
+-----------+-------------+
2 rows in set (0.99 sec)

mysql>
```